



STATUTORY EMISSIONS AUDIT & CIRCULAR FEASIBILITY BOOKLET

Industrial Decarbonization & Circularity Roadmap

A comprehensive engineering dossier isolating process leakage points, quantified Scope 1–3 emissions, and verified circular intervention payback models for capital approval.

TARGET INDUSTRIAL ESTABLISHMENT

Facility / Unit Name

Plastic manufacturing

Geographic Site

Industrial Facility Site

Consent to Operate (CTO) Reg ID

SPCB/CTO-2026/9530

Pollution Control Category

Orange Category

BASELINE MONTHLY FOOTPRINT

150.29 tCO₂e

Total Scope 1–3 emissions

RECOVERABLE CARBON

114.36 tCO₂e

76% monthly emissions reduction

ANNUAL OPEX CASH SAVED

₹31.1 Lakh

Recurring operating cash recovery

★ TANGIBLE CLIMATE EQUIVALENTS (EPA GHG STANDARD)

Implementing this action plan abates **114.36 tCO₂e/mo**, equivalent to taking **25 cars off the road for a year**, planting **5,253 trees**, or saving **266 barrels of crude oil** from industrial extraction.

⚠️ Top Emission Leak Points (Ranked by 80/20 Pareto Magnitude)

RANK	ACTIVITY / FUEL SOURCE	SCOPE	MONTHLY CARBON	SHARE %	SEVERITY TIER
#1	Virgin Plastic Pellets	Scope 3	123.0 tCO ₂ e	81.84%	Critical Hotspot
#2	Grid Electricity	Scope 2	14.2 tCO ₂ e	9.45%	High Leak
#3	Color Additives	Scope 3	7.0 tCO ₂ e	4.66%	High Leak
#4	Packaging Material	Scope 3	4.75 tCO ₂ e	3.16%	High Leak
#5	Diesel Fuel	Scope 1	1.34 tCO ₂ e	0.89%	High Leak

🔄 Pre-Engineered Circular Solutions & Capital Amortization

INTERVENTION ALTERNATIVE	TARGET HOTSPOT	CO ₂ CUT	EST. CAPEX	MONTHLY OPEX SAVED	PAYBACK
Recycled Polymer Pellets (PCR)	Virgin Plastic Pellets	-50.39%	₹3,30,990	+₹2,25,000/mo	1.5 Months
Onsite Rooftop Solar	Grid Electricity	-93.66%	₹13,34,913	+₹8,333/mo	160.2 Months
Bio-Carrier Masterbatch	Color Additives & Pigments	-60.71%	₹40,613	+₹12,500/mo	3.2 Months
Recycled Corrugated Packaging	Packaging Material	-8.58%	₹44,263	+₹13,333/mo	3.3 Months

🛡️ REGULATORY ASSURANCE & STATUTORY ALIGNMENT

This engineering plan complies with **Central Pollution Control Board (CPCB)** charter norms, **SEBI BRSR Core Indicator 1–2** reporting guidelines, **ISO 14064-1** carbon accounting protocols, and **EU CBAM Annex IV** embedded emissions formulas.

Prepared by:
Kavin Jindal

EcoLeak Engineering Engine · Verified Local Execution

Factory Compliance Authorization:

Chief Sustainability Officer / Operations Director